Lecture 13: Laplace Expansion

MATH203 Applied Linear Algebra

Laplace Expansion

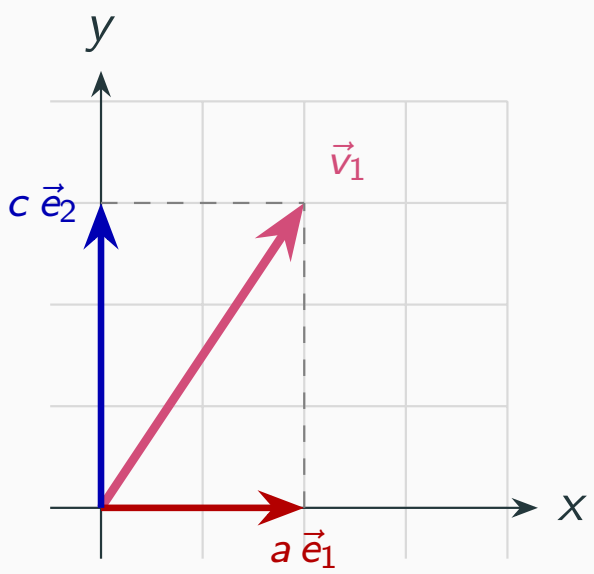
Can we break a **big** determinant
into **smaller** determinants?

In Lecture 12 we computed det by cross-filling.

Today: a different decomposition — expand **one column** at a time.

The Key Idea: Decompose One Column

Any column is a sum of scaled basis vectors:



$$\begin{pmatrix} a \\ c \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

By **multilinearity** (P1):

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} =$$

$$a \det \begin{pmatrix} 1 & b \\ 0 & d \end{pmatrix} + c \det \begin{pmatrix} 0 & b \\ 1 & d \end{pmatrix}$$

Each piece: column 1 replaced by $\vec{e}_i$ — a **simpler** determinant!

The 2×2 Case: Nothing New

Compute each piece:

$$\det \begin{pmatrix} 1 & b \\ 0 & d \end{pmatrix} = 1 \cdot d - 0 \cdot b = \mathbf{d}$$

$$\det \begin{pmatrix} 0 & b \\ 1 & d \end{pmatrix} = 0 \cdot d - 1 \cdot b = \mathbf{-b}$$

Combine:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = a \cdot \mathbf{d} + c \cdot (\mathbf{-b}) = ad - bc \quad \checkmark$$

The familiar formula is just **Laplace expansion along column 1**!

The 3×3 Case

Take the matrix from Lecture 12 (det $= 5$ by cross-filling):

$$A = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 3 \end{pmatrix}$$

Decompose **column 1** into basis vectors:

$$\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 1 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Apply **multilinearity** (P1) in column 1 ...

Expanding Column 1

P1 gives three terms — each replaces column 1 by $\vec{e}_i$:

$$\det(A) = 2\det\begin{pmatrix} 1 & 1 & 2 \\ 0 & 3 & 1 \\ 0 & 1 & 3 \end{pmatrix} + 1\det\begin{pmatrix} 0 & 1 & 2 \\ 1 & 3 & 1 \\ 0 & 1 & 3 \end{pmatrix} + 2\det\begin{pmatrix} 0 & 1 & 2 \\ 0 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix}$$

Each piece: a 3×3 determinant with **one** 1 and **rest** 0 in column 1

We give these pieces a name . . .

Algebraic Cofactor A_{ij}

Definition

The **algebraic cofactor** A_{ij} is the determinant of A with **column** j replaced by $\vec{e}_i$:

$$A_{ij} = \det\left(\vec{v}_1, \ldots, \underbrace{\vec{e}_i}_{\text{column } j}, \ldots, \vec{v}_n\right)$$

Recipe: to write down A_{ij}, take A and:

1. Put **1** at position (i, j)
2. Put **0** everywhere else in column j
3. Leave all other columns **unchanged**

Then compute the determinant of the result.

Computing A_{11}

$$A_{11} = \det \begin{pmatrix} 1 & 1 & 2 \\ 0 & 3 & 1 \\ 0 & 1 & 3 \end{pmatrix}$$

Column 1 is $\vec{e}_1$: the 1 sits **on top**, with 0's below.

This is **block upper triangular** (Lecture 12, §5):

$$\det \begin{pmatrix} 1 & 1 & 2 \\ 0 & 3 & 1 \\ 0 & 1 & 3 \end{pmatrix} \quad = \quad 1 \times \det \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} = 9 - 1 = \mathbf{8}$$

Visual: cross out the row and column where the 1 sits.

The remaining 2×2 block gives the answer.

Computing A_{21} and A_{31}

A_{21}: column 1 is $\vec{e}_2$ — the 1 is in **row 2**, not on top.

Swap rows $1 \leftrightarrow 2$ (sign flips!), then block triangular:

$$A_{21} = \det \begin{pmatrix} 0 & 1 & 2 \\ 1 & 3 & 1 \\ 0 & 1 & 3 \end{pmatrix} \overset{r_1 \leftrightarrow r_2}{=} -\det \begin{pmatrix} 1 & 3 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 3 \end{pmatrix} = -(3-2) = \mathbf{-1}$$

A_{31}: column 1 is $\vec{e}_3$ — the 1 is in **row 3**.

Swap rows $1 \leftrightarrow 3$ (sign flips!), then block triangular:

$$A_{31} = \det \begin{pmatrix} 0 & 1 & 2 \\ 0 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix} \overset{r_1 \leftrightarrow r_3}{=} -\det \begin{pmatrix} 1 & 1 & 3 \\ 0 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix} = -(6-1) = \mathbf{-5}$$

The Pattern: Cross Out and Compute

Cross out row i and column j where the 1 sits. The remaining block gives the answer (up to sign from row swaps):

A_{11}: cross row 1, col 1

$$\begin{pmatrix} \cancel{1} & \cancel{1} & \cancel{2} \\ \cancel{0} & 3 & 1 \\ \cancel{0} & 1 & 3 \end{pmatrix}$$

$$+\det\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} = \mathbf{8}$$

A_{21}: cross row 2, col 1

$$\begin{pmatrix} \cancel{0} & 1 & 2 \\ \cancel{1} & \cancel{3} & \cancel{1} \\ \cancel{0} & 1 & 3 \end{pmatrix}$$

$$-\det\begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix} = \mathbf{-1}$$

A_{31}: cross row 3, col 1

$$\begin{pmatrix} \cancel{0} & 1 & 2 \\ \cancel{0} & 3 & 1 \\ \cancel{1} & \cancel{1} & \cancel{3} \end{pmatrix}$$

$$-\det\begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix} = \mathbf{-5}$$

Putting It Together

$$\det(A) = \underbrace{2}_{a_{11}} \cdot \underbrace{8}_{A_{11}} + \underbrace{1}_{a_{21}} \cdot \underbrace{(-1)}_{A_{21}} + \underbrace{2}_{a_{31}} \cdot \underbrace{(-5)}_{A_{31}}$$

$$= 16 - 1 - 10 = \mathbf{5} \quad \checkmark$$

Matches cross-filling from Lecture 12!

Theorem (Laplace Expansion along column j)

$$\boxed{\det(A) = \sum_{i=1}^{n} a_{ij} \cdot A_{ij} = a_{1j}\, A_{1j} + a_{2j}\, A_{2j} + \cdots + a_{nj}\, A_{nj}}$$

Since $\det(A^T) = \det(A)$, the same formula works for **rows**: fix row i, sum over j.

Strategy: Expand Along Zeros

$$\det \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 0 \\ 2 & 1 & 3 \end{pmatrix}$$

Strategy: Expand Along Zeros

$$\det \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 0 \\ 2 & 1 & 3 \end{pmatrix}$$

Column 3 has **two zeros** — expand along it! Only one term survives:

$$= \underbrace{0}_{a_{13}} \cdot A_{13} + \underbrace{0}_{a_{23}} \cdot A_{23} + \underbrace{3}_{a_{33}} \cdot A_{33} = 3 \cdot A_{33}$$

A_{33}: put 1 at $(3,3)$, zeros elsewhere in col 3. Block triangular:

$$A_{33} = \det \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 0 \\ 2 & 1 & 1 \end{pmatrix} = \det \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} = 5$$

$$\det = 3 \times 5 = \mathbf{15}$$

Practice

Compute by Laplace expansion (choose a good row or column!):

1. $\det \begin{pmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 1 \end{pmatrix} =$

2. $\det \begin{pmatrix} 1 & 2 & 0 & 0 \\ 3 & 1 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & 4 \end{pmatrix} =$

3. For the 4×4 matrix above, which is faster: Laplace or block triangular?

The Orthogonality Trick

Recap: Column 1 Entries $\times$ Column 1 Cofactors

$$A = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 3 \end{pmatrix}$$

Column 1 entries: 2, 1, 2.

Column 1 cofactors (replace column 1 by $\vec{e}_i$, from §1):

$$A_{11} = \det \begin{pmatrix} 1 & 1 & 2 \\ 0 & 3 & 1 \\ 0 & 1 & 3 \end{pmatrix} = 8, \quad A_{21} = \det \begin{pmatrix} 0 & 1 & 2 \\ 1 & 3 & 1 \\ 0 & 1 & 3 \end{pmatrix} = -1,$$

$$A_{31} = \det \begin{pmatrix} 0 & 1 & 2 \\ 0 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix} = -5$$

Column 1 entries $\times$ column 1 cofactors = Laplace expansion = $\det(A)$:

$$\underbrace{2}_{a_{11}} \cdot \underbrace{8}_{A_{11}} + \underbrace{1}_{a_{21}} \cdot \underbrace{(-1)}_{A_{21}} + \underbrace{2}_{a_{31}} \cdot \underbrace{(-5)}_{A_{31}} = \mathbf{5} = \det(A) \quad \checkmark$$

What if we keep the **same cofactors**

$$A_{11} = 8, \quad A_{21} = -1, \quad A_{31} = -5$$

but multiply them by entries from a **different** column of A?

Column 2 Entries $\times$ Column 1 Cofactors

Take column 2 entries from $A = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 3 \end{pmatrix}$, multiply by the **same** column 1 cofactors:

$$\underbrace{1}_{a_{12}} \cdot \underbrace{8}_{A_{11}} + \underbrace{3}_{a_{22}} \cdot \underbrace{(-1)}_{A_{21}} + \underbrace{1}_{a_{32}} \cdot \underbrace{(-5)}_{A_{31}} = 8 - 3 - 5 = \mathbf{0}$$

Why? This sum is the Laplace expansion of a **fake matrix** — copy column 2 into column 1:

$$A = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 3 \end{pmatrix} \xrightarrow{\text{copy col 2}} \underbrace{\begin{pmatrix} 1 & 1 & 2 \\ 3 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix}}_{\text{same!}}$$

$\det = 5$ $\qquad$ $\det = \mathbf{0}$

Two identical blue columns $\Rightarrow$ P2 $\Rightarrow$ $\det = \mathbf{0}$.

Column 3 Entries $\times$ Column 1 Cofactors

Take column 3 entries from $A = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 3 \end{pmatrix}$, multiply by the **same** column 1 cofactors:

$$\underbrace{2}_{a_{13}} \cdot \underbrace{8}_{A_{11}} + \underbrace{1}_{a_{23}} \cdot \underbrace{(-1)}_{A_{21}} + \underbrace{3}_{a_{33}} \cdot \underbrace{(-5)}_{A_{31}} = 16 - 1 - 15 = \mathbf{0}$$

Why? Copy column 3 into column 1:

$$A = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 3 \end{pmatrix} \xrightarrow{\text{copy col 3}} \begin{pmatrix} \overbrace{2 \quad 1}^{\text{same}} & \overbrace{2}^{\text{same}} \\ 1 \quad 3 & 1 \\ 3 \quad 1 & 3 \end{pmatrix}$$

$\det = 5$ $\qquad$ $\det = \mathbf{0}$

Two identical green columns (col 1 and col 3) $\Rightarrow$ P2 $\Rightarrow$ $\det = \mathbf{0}$.

Summary: Right Column vs. Wrong Column

All three cases with column 1 cofactors ($A_{11}=8,\ A_{21}=-1,\ A_{31}=-5$):

	entries $\times$ cofactors	result
Col 1 entries: $2, 1, 2$	$= \det(A)$ by Laplace	$= \mathbf{5}$
Col 2 entries: $1, 3, 1$	$= \det(\text{fake: two blue cols})$	$= \mathbf{0}$
Col 3 entries: $2, 1, 3$	$= \det(\text{fake: two green cols})$	$= \mathbf{0}$

Right column's entries give $\det(A)$. **Wrong** column's entries give 0.

Wrong column creates a **repeated column** in the fake matrix $\Rightarrow$ P2 $\Rightarrow$ det $= 0$

The Cofactor Orthogonality

Theorem

$$\sum_{i=1}^{n} a_{ik} \cdot A_{ij} = \begin{cases} \det(A) & \text{if } k = j \quad \text{(right column)} \\ 0 & \text{if } k \neq j \quad \text{(wrong column)} \end{cases}$$

For our 3×3 example ($\det A = 5$):

	col 1 cofactors	col 2 cofactors	col 3 cofactors
col 1 entries	5	0	0
col 2 entries	0	5	0
col 3 entries	0	0	5

Diagonal $= \det(A)$, off-diagonal $= 0$. This is $\det(A) \cdot I$!

Adjugate Matrix

2×2 Adjugate: Each Entry Is a Cofactor

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad \det(A) = 5.$$

Four cofactors (replace column j by $\vec{e}_i$, take det):

$$A^* = \begin{pmatrix} A_{11} & A_{21} \\ A_{12} & A_{22} \end{pmatrix} = \begin{pmatrix} \det\begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix} & \det\begin{pmatrix} 0 & 1 \\ 1 & 3 \end{pmatrix} \\ \det\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} & \det\begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}$$

Shortcut: swap diagonal, negate off-diagonal:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \to A^* = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Multiply: $A^* \cdot A$

Row i of A^* = cofactors of col i. Column j of A = entries of col j.

$$\underbrace{\begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}}_{A^*} \cdot \underbrace{\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}}_{A} = ?$$

entry	dot product
$(1,1)$: col 1 cofactors $\times$ col 1 entries	$3 \cdot 2 + (-1) \cdot 1$
$(1,2)$: col 1 cofactors $\times$ col 2 entries	$3 \cdot 1 + (-1) \cdot 3$
$(2,1)$: col 2 cofactors $\times$ col 1 entries	$(-1) \cdot 2 + 2 \cdot 1$
$(2,2)$: col 2 cofactors $\times$ col 2 entries	$(-1) \cdot 1 + 2 \cdot 3$

Same color = matching $\rightarrow$ Laplace? **Mixed color** = mismatch $\rightarrow$???

Diagonal Entry $(1,1)$: Laplace Along Col 1

Col 1 cofactors $\times$ col 1 entries:

$$2 \cdot \underbrace{\det\begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}}_{A_{11}} + 1 \cdot \underbrace{\det\begin{pmatrix} 0 & 1 \\ 1 & 3 \end{pmatrix}}_{A_{21}}$$

Coefficients 2, 1 go back into col 1 $\implies$ Laplace expansion:

$$= \det\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} = 2 \cdot 3 - 1 \cdot 1 = \mathbf{5} = \det(A) \checkmark$$

Diagonal Entry $(2,2)$: Laplace Along Col 2

Col 2 cofactors $\times$ col 2 entries:

$$1 \cdot \underbrace{\det\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}}_{A_{12}} + 3 \cdot \underbrace{\det\begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}}_{A_{22}}$$

Coefficients 1, 3 go back into col 2 $\implies$ Laplace expansion:

$$= \det\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} = 2 \cdot 3 - 1 \cdot 1 = \mathbf{5} = \det(A) \checkmark$$

Off-Diagonal Entry $(1,2)$: Wrong Column

Col 1 cofactors $\times$ col 2 entries:

$$1 \cdot \underbrace{\det\begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}}_{A_{11}} + 3 \cdot \underbrace{\det\begin{pmatrix} 0 & 1 \\ 1 & 3 \end{pmatrix}}_{A_{21}}$$

Coefficients 1, 3 go into col 1 position $\implies$

$$= \det\begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix}$$

But col 2 is also $(1,3)^T \implies$ **two identical columns** $\implies$ P2 $\implies = \mathbf{0}$

Off-Diagonal Entry $(2,1)$: Wrong Column Again

Col 2 cofactors $\times$ col 1 entries:

$$2 \cdot \underbrace{\det\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}}_{A_{12}} + 1 \cdot \underbrace{\det\begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}}_{A_{22}}$$

Coefficients 2, 1 go into col 2 position $\implies$

$$= \det\begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}$$

But col 1 is also $(2,1)^T \implies$ **two identical columns** $\implies$ P2 $\implies = \mathbf{0}$

$$\therefore \quad A^* \cdot A = \begin{pmatrix} \mathbf{5} & 0 \\ 0 & \mathbf{5} \end{pmatrix} = \det(A) \cdot I_2 \ \checkmark$$

Shortcut: The Cross-Out Rule

Computing each cofactor by row-swaps is tedious. A shortcut:

$$\boxed{A_{ij} = \underbrace{(-1)^{i+j}}_{\text{checkerboard sign}} \cdot \underbrace{\det\big(\text{delete row } i,\ \text{col } j \text{ from } A\big)}_{\text{minor } M_{ij}}}$$

Sign pattern: $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$

Verify A_{21}: cross out row 2, col 1:

$$A_{21} = \underbrace{(-1)^{2+1}}_{-1} \cdot \underbrace{\det\begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}}_{3-2=1} = -1 \quad \checkmark$$

Computing Column 2 Cofactors

$$A = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 3 \end{pmatrix}$$

Cross out col 2 and the indicated row:

$$A_{12}: \underbrace{(-1)^{1+2}}_{-1} \cdot \det\begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} = (-1) \cdot \underbrace{(3-2)}_{1} = -1$$

$$A_{22}: \underbrace{(-1)^{2+2}}_{+1} \cdot \det\begin{pmatrix} 2 & 2 \\ 2 & 3 \end{pmatrix} = (+1) \cdot \underbrace{(6-4)}_{2} = 2$$

$$A_{32}: \underbrace{(-1)^{3+2}}_{-1} \cdot \det\begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} = (-1) \cdot \underbrace{(2-2)}_{0} = 0$$

Computing Column 3 Cofactors

$$A = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 3 \end{pmatrix}$$

Cross out col 3 and the indicated row:

$$A_{13}: \underbrace{(-1)^{1+3}}_{+1} \cdot \det\begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix} = (+1) \cdot \underbrace{(1-6)}_{-5} = -5$$

$$A_{23}: \underbrace{(-1)^{2+3}}_{-1} \cdot \det\begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} = (-1) \cdot \underbrace{(2-2)}_{0} = 0$$

$$A_{33}: \underbrace{(-1)^{3+3}}_{+1} \cdot \det\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} = (+1) \cdot \underbrace{(6-1)}_{5} = 5$$

Packing Cofactors: The Adjugate A^*

All 9 cofactors:

	col 1	col 2	col 3
row 1	8	-1	-5
row 2	-1	2	0
row 3	-5	0	5

The **adjugate** is the **transpose** of the cofactor table:

$$A^* := (\text{cofactor matrix})^T, \qquad (A^*)_{ij} = A_{ji}$$

$$A^* = \begin{pmatrix} 8 & -1 & -5 \\ -1 & 2 & 0 \\ -5 & 0 & 5 \end{pmatrix} \longleftarrow \text{row } i \text{ of } A^* = \text{cofactors of col } i \text{ of } A$$

3×3 **Verification:** $A^* \cdot A = \det(A) \cdot I$

Theorem

$$\boxed{A^* \cdot A = \det(A) \cdot I_n}$$

Row 1 of A^* = col 1 cofactors $= (8, -1, -5)$, dot with each column of A:

$$(8, -1, -5) \cdot (2, 1, 2) = \underbrace{16 - 1 - 10}_{=5} = \det(A) \checkmark \qquad \text{(Laplace along col 1)}$$

$$(8, -1, -5) \cdot (1, 3, 1) = \underbrace{8 - 3 - 5}_{=0} \checkmark \qquad \text{(col 1 cofactors} \times \text{col 2 entries} \Rightarrow 0)$$

$$(8, -1, -5) \cdot (2, 1, 3) = \underbrace{16 - 1 - 15}_{=0} \checkmark \qquad \text{(col 1 cofactors} \times \text{col 3 entries} \Rightarrow 0)$$

Rows 2, 3 of A^*: same pattern. Result: $A^* \cdot A = 5\, I_3 \checkmark$

2×2 Adjugate: Swap & Negate

$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Using the cross-out rule:

$$A_{11} = \underbrace{(+)}_{(-1)^{1+1}} d = d, \qquad A_{12} = \underbrace{(-)}_{(-1)^{1+2}} c = -c$$

$$A_{21} = \underbrace{(-)}_{(-1)^{2+1}} b = -b, \qquad A_{22} = \underbrace{(+)}_{(-1)^{2+2}} a = a$$

Cofactor matrix $\begin{pmatrix} d & -c \\ -b & a \end{pmatrix} \xrightarrow{\text{transpose}}$ Adjugate $A^* = \begin{pmatrix} d & \textbf{-b} \\ \textbf{-c} & a \end{pmatrix}$

Mnemonic: **swap** diagonal entries, **negate** off-diagonal entries.

$$\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} ad - bc & 0 \\ 0 & ad - bc \end{pmatrix} = (ad - bc)\, I_2 \checkmark$$

Inverse via Adjugate

The Inverse Formula

From $A^* \cdot A = \det(A) \cdot I$, divide both sides by $\det(A)$:

Theorem

If $\det(A) \neq 0$:

$$\boxed{A^{-1} = \frac{1}{\det(A)} A^*}$$

Our example: $\det(A) = 5 \neq 0$, so

$$A^{-1} = \frac{1}{5} \begin{pmatrix} 8 & -1 & -5 \\ -1 & 2 & 0 \\ -5 & 0 & 5 \end{pmatrix}$$

When Is a Matrix Invertible?

Theorem

$$\boxed{A \text{ is invertible} \iff \det(A) \neq 0}$$

Proof ($\Leftarrow$): If $\det(A) \neq 0$, we just built the inverse:

$$A^{-1} = \frac{1}{\det(A)} A^*$$

Proof ($\Rightarrow$): If A is invertible, then $AA^{-1} = I$, so

$$\det(A)\det(A^{-1}) = \det(I) = 1 \implies \det(A) \neq 0$$

Contrapositive: $\det(A) = 0 \implies A$ is **not invertible** (columns are linearly dependent)

2×2 Inverse Formula

Theorem

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Example: $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}^{-1}$

$$= \frac{1}{\underbrace{2 \cdot 3 - 1 \cdot 1}_{5}} \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} \frac{3}{5} & -\frac{1}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{pmatrix}$$

Verify: $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 3/5 & -1/5 \\ -1/5 & 2/5 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ ✓

Cramer's Rule

From Adjugate to Cramer

For $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} p \\ q \end{pmatrix}$, the inverse formula gives:

$$x = \frac{dp - bq}{ad - bc}, \qquad y = \frac{aq - cp}{ad - bc}$$

Look at the numerator of x:

$$\underbrace{dp - bq}_{\text{numerator}} = \det \begin{pmatrix} p & b \\ q & d \end{pmatrix} \quad \longleftarrow \quad A \text{ with col 1 replaced by } \begin{pmatrix} p \\ q \end{pmatrix}$$

Look at the numerator of y:

$$\underbrace{aq - cp}_{\text{numerator}} = \det \begin{pmatrix} a & p \\ c & q \end{pmatrix} \quad \longleftarrow \quad A \text{ with col 2 replaced by } \begin{pmatrix} p \\ q \end{pmatrix}$$

Pattern: $x_i = \dfrac{\det(A_i)}{\det(A)}$ where A_i = replace column i by b!

Cramer's Rule

Theorem (Cramer's Rule)

If $\det(A) \neq 0$ and $Ax = b$, then

$$\boxed{x_i = \frac{\det(A_i)}{\det(A)}}$$

where $A_i = A$ with **column i replaced by b**.

Visually for 3×3: which column gets replaced?

$$A_1 = \begin{pmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{pmatrix}, \quad A_2 = \begin{pmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{pmatrix}, \quad A_3 = \begin{pmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{pmatrix}$$

Why? Cofactors of col i depend only on other columns — same in A and A_i.

So $\sum_k A_{ki} b_k$ = Laplace expansion of $\det(A_i)$ along col i.

Example: Cramer's Rule

$$\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}, \quad \det(A) = \underbrace{6 - 1}_{5}.$$

$$x = \frac{\det \begin{pmatrix} 5 & 1 \\ 5 & 3 \end{pmatrix}}{\det(A)} = \frac{\overbrace{15 - 5}^{10}}{5} = 2$$

$$y = \frac{\det \begin{pmatrix} 2 & 5 \\ 1 & 5 \end{pmatrix}}{\det(A)} = \frac{\overbrace{10 - 5}^{5}}{5} = 1$$

Check: $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$ ✓

Practice

1. Use Cramer's Rule to solve:

$$\begin{pmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 7 \\ 6 \\ 8 \end{pmatrix}$$

   Hint: $\det(A) = 5$. Use $x_i = \frac{1}{\det A}(\text{row } i \text{ of } A^*) \cdot b$ — you know all the cofactors!

2. Find $\begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}^{-1}$ using the adjugate formula.

3. True or False: if $\det(A) = 0$, then $Ax = b$ has no solution.

Summary

1. **Cross-out rule**: $A_{ij} = (-1)^{i+j} M_{ij}$
2. **Adjugate**: $A^* = (\text{cofactor matrix})^T, \quad A \cdot A^* = \det(A) \cdot I$
3. **Inverse**: $A^{-1} = \dfrac{1}{\det A} A^* \quad (\text{when } \det A \neq 0)$
4. **Cramer**: $x_i = \dfrac{\det(A_i)}{\det(A)}$

All from **two properties**: multilinearity (P1) + alternating (P2)!